radically so that the force transmitted by the candidate gravitons would have the familiar, known strength of gravity. Since gravity is an extremely weak force[32] and since, it turns out, the longer the string the *stronger* the force transmitted, Schwarz and Scherk found that strings needed to be extremely tiny to transmit a force with gravity's feeble strength; they needed to be about the Planck length in size, a hundred billion billion times smaller than previously envisioned. So small, doubters wryly noted, that there was no equipment that would be able to see them, which meant that the theory could not be tested experimentally.[10]

By contrast, the 1970s witnessed one success after another for the more conventional, non-string-based theories, formulated with point particles and fields. Theorists and experimenters alike had their heads and hands full of concrete ideas to investigate and predictions to test. Why turn to speculative string theory when there was so much exciting work to be done within a tried-and-true framework? In much the same vein, although physicists knew in the backs of their minds that the problem of merging gravity and quantum mechanics remained unsolved using conventional methods, it was not a problem that commanded attention. Almost everyone acknowledged that it was an important issue and would need to be addressed one day, but with the wealth of work still to be done on the nongravitational forces, the problem of quantizing gravity was pushed to a barely burning back burner. And, finally, in the mid to late 1970s, string theory was far from having been completely worked out. Containing a candidate for the graviton was a success, but many conceptual and technical issues had yet to be addressed. It seemed thoroughly plausible that the theory would be unable to surmount one or more of these issues, so working on string theory meant taking a considerable risk. Within a few years, the theory might be dead.

Schwarz remained resolute. He believed that the discovery of string theory, the first plausible approach for describing gravity in the language of quantum mechanics, was a major breakthrough. If no one wanted to listen, fine. He would press on and develop the theory, so that when people were ready to pay attention, string theory would be that much further along. His determination proved prescient.